



1. Perform data cleaning operations and visualize the data

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Sign here to indicate that you have read all relevant material provided /available on Moodle while performing and writing this experiment		Sign:	

Late Submission Details (if any)

Reason(s) of late submission	Date of practical performance	Date of practical submission

References used

1	Name and author of reference book(s) with page nos.	
2	Name and roll nos. of the peers whose help you have taken (if any)	

Rubrics for assessment of Experiment:

Indicator	Poor	Average	Good
Timeliness Maintains Experiment deadline (3)	Experiment not done (0)	One or More than One week late (1-2)	Maintains deadline (3)
Completeness and neatness Complete all parts of Experiment (3)	N/A	< 80% complete (1-2)	100% complete (3)
Originality Extent of plagiarism (2)	Copied it from someone else (0)	At least try to implement but could not succeed (1)	Implemented (2)
Knowledge In depth knowledge of the Experiment (2)	Unable to answer any questions (0)	Unable to answer few questions (1)	Able to answer all questions (2)

Assessment Marks:

Timeliness	
Completeness and neatness	
Originality	
Knowledge	
Total	

Signature of Teacher with date

1. Course, Subject & Experiment Details

Course & Branch	T.E. (ECS)	Estimated Time	02 Hours Per Week
Current Semester	Semester VI	Subject Name	DWM
Chapter No. & Unit	-	Chapter Title	-
Experiment Type	Software Performance	Subject Code	25PCC13EC18

2. Aim & Objective of Experiment

1. Apply and analyze data handling techniques by loading, preprocessing, and visualizing datasets using Pandas, Matplotlib, and Seaborn (CO1)

3. Expected Outcome of Experiment

1. To study data cleaning operations and visualize the data

4. References

5. Batch and Group No:

Dataset Sr. No

6. URL of the dataset:

<https://www.bing.com/search?q=uci+dataset+adult&form=ANNTTH1&refig=6987201d6ce14b4b8ffd441736405246&pc=W037>

7. Description of the dataset and its snippet

9. Code snippet and its explanation

(Write sample code and explain it)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

1. Define columns based on the variable list provided

```
cols = [
    "age", "workclass", "fnlwgt", "education", "education-num",
```

```
"marital-status", "occupation", "relationship", "race", "sex",  
"capital-gain", "capital-loss", "hours-per-week", "native-country", "income"  
]  
  
# 2. Load dataset with headers and trim whitespace  
df = pd.read_csv("C:\\Users\\universe\\adult.csv", names=cols, skipinitialspace=True)  
  
# 3. Replace '?' with NaN (as specified in the dataset documentation)  
df.replace('?', np.nan, inplace=True)  
  
# 4. Data structure and missing values  
print(df.info())  
print(df.isnull().sum())  
  
# 5. Descriptive statistics  
print(df.describe())  
  
# 6. Remove duplicates  
df.drop_duplicates(inplace=True)  
  
# 7. Handle missing numerical values using mean  
num_cols = df.select_dtypes(include=[np.number]).columns  
for col in num_cols:  
    df[col] = df[col].fillna(df[col].mean())  
  
# 8. Handle missing categorical values using mode  
cat_cols = df.select_dtypes(include=['object']).columns  
for col in cat_cols:  
    df[col] = df[col].fillna(df[col].mode()[0])  
  
# 9. Select rows where age > average (This will now work without KeyError)  
avg_age = df['age'].mean()  
greater_than_avg = df[df['age'] > avg_age]  
print(f"Entries older than average ({avg_age:.2f}): {len(greater_than_avg)}")  
  
# 10. Sort by age  
df_sorted = df.sort_values(by='age')  
  
# 11. Group by education and calculate average hours-per-week  
grouped_data = df.groupby('education')['hours-per-week'].mean()  
print(grouped_data)  
  
# 12-15. Visualizations  
plt.figure(figsize=(10, 6))  
df['age'].hist(bins=20, color='skyblue', edgecolor='black')  
plt.title('Age Distribution')  
plt.show()  
  
# Boxplot for Outliers  
plt.figure(figsize=(8, 5))
```

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```
plt.boxplot(df['hours-per-week'])  
plt.title('Hours per Week Outliers')  
plt.show()
```

(Also attach .ipynb file while uploading this write-up)

10 All output screenshots with explanations

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    32561 non-null  int64
1   workclass              30725 non-null  object
2   fnlwgt                 32561 non-null  int64
3   education              32561 non-null  object
4   education-num          32560 non-null  float64
5   marital-status         32561 non-null  object
6   occupation             30718 non-null  object
7   relationship           32560 non-null  object
8   race                   32559 non-null  object
9   sex                    32561 non-null  object
10  capital-gain           32560 non-null  float64
11  capital-loss           32561 non-null  int64
12  hours-per-week         32561 non-null  int64
13  native-country         31974 non-null  object
14  income                 32561 non-null  object
dtypes: float64(2), int64(4), object(9)
memory usage: 3.7+ MB
None
age                    0
workclass              1836
fnlwgt                 0
education              0
education-num          1
marital-status         0
occupation             1843
relationship           1
race                   2
sex                    0
capital-gain           1
capital-loss           0
hours-per-week         0
native-country         587
income                 0
dtype: int64

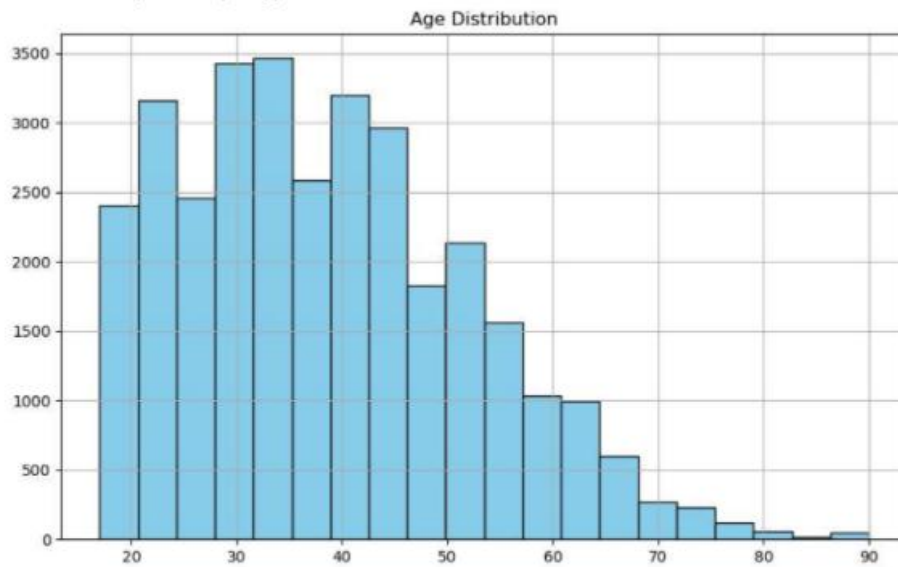
count  age          fnlwgt  education-num  capital-gain  capital-loss \
mean    38.581647  1.897784e+05   10.080498     1077.522912    87.303830
std     13.640433  1.055500e+05    2.572551     7385.370537   402.960219
min     17.000000  1.228500e+04    1.000000      0.000000      0.000000
25%     28.000000  1.178270e+05    9.000000      0.000000      0.000000
50%     37.000000  1.783560e+05   10.000000      0.000000      0.000000
75%     48.000000  2.370510e+05   12.000000      0.000000      0.000000
max     90.000000  1.484705e+06   16.000000  99999.000000  4356.000000

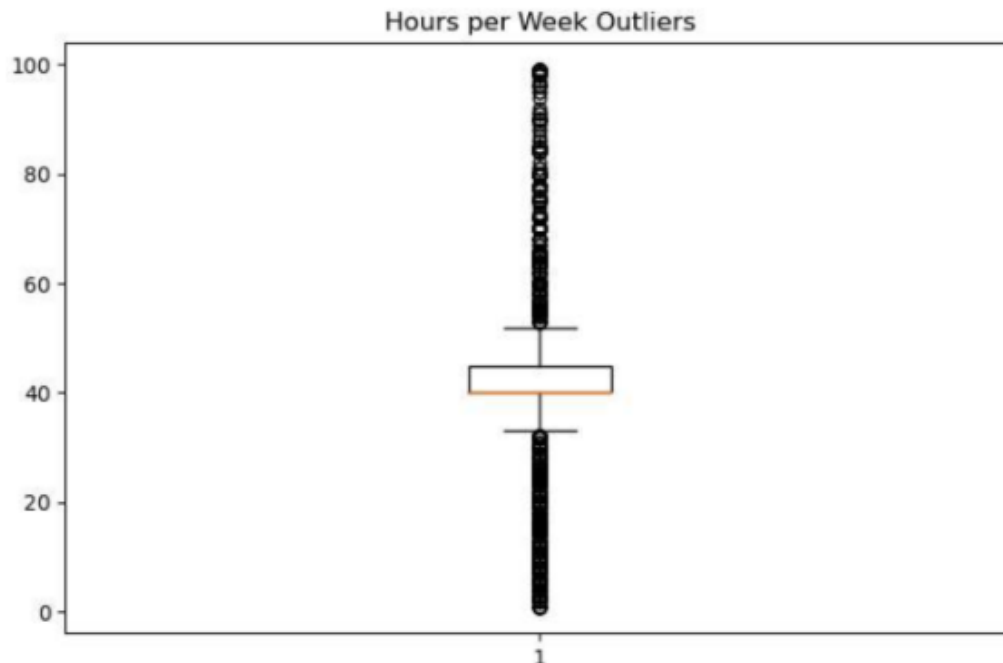
hours-per-week
count    32561.000000
mean      40.437456
std       12.347429
min        1.000000
25%       40.000000
50%       40.000000
75%       45.000000
max       99.000000
Entries older than average (38.59): 15045
education
```

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10th	37.052519
11th	33.925957
12th	35.780600
1st-4th	38.234940
5th-6th	38.849398
7th-8th	39.365891
9th	38.044747
Assoc-acdm	40.504217
Assoc-voc	41.610709
Bachelors	42.614048
Doctorate	46.973366
HS-grad	40.574995
Masters	43.832753
Preschool	36.380000
Prof-school	47.425347
Some-college	38.868855

Name: hours-per-week, dtype: float64





The output indicates that the Adult Census Income dataset contains 32,561 records and 15 attributes. From the statistical summary, the average age of individuals is approximately 38.59 years, with a minimum age of 17 and a maximum of 90, showing a wide age range. The average working hours per week is about 40.43 hours, with most individuals working between 40 and 45 hours, while a few outliers work up to 99 hours per week. The missing value analysis shows that attributes like workclass (1836 missing values), occupation (1843 missing values), and native-country (587 missing values) initially contained missing data, which were handled using mean and mode imputation. The grouping result shows that individuals with higher education levels such as Doctorate (≈ 47 hours/week) and Professional school (≈ 47.4 hours/week) work more hours on average compared to those with lower education levels. The histogram confirms that most individuals fall in the 25–50 age range, and the boxplot highlights the presence of outliers in working hours.

5. Explain what you have learned from this experiment in your own words.

In this experiment, I learned how to load and preprocess a real-world dataset using Python. I understood how to handle missing values and clean the data properly. I learned to compute statistical measures to analyze the dataset. This experiment showed the importance of data analysis before applying machine learning models.

6. Post Lab exercise

1. What other statistical values can be obtained for the given data (other than mean, std. deviation, min and max)

Apart from mean, standard deviation, minimum and maximum, the following statistical values can be obtained:

- Median
- Mode

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- Variance
- Range
- Quartiles (Q1, Q2, Q3)
- Percentiles

2. How add an attribute in the data frame, which depends on an attribute of the main data?

A new attribute can be added by creating a new column derived from an existing column using arithmetic operations or conditional logic.

```
df['age_group'] = df['age'].apply(  
    lambda x: 'Young' if x < 30 else 'Adult'  
)
```
